

TRIADS: Tiny Recursive Information-Attention with Deep Supervision

A Study in Parameter-Efficient Recursive Architectures for Materials

Property Prediction Across Six Matbench Benchmarks with <250K Parameters

Rudra Tiwari

Independent Researcher

[GitHub](#) | [Hugging Face](#) | [Website](#)

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Abstract

In small-data materials ML, the central mechanism in TRIADS is the coupling of recursive attention with per-cycle deep supervision. On `matbench_expt_is_metal`, a **44K**-parameter TRIADS model reaches **0.9644** ROC-AUC and outperforms GPTChem (**0.8965**, $\gg$ 1B parameters) by **6.79** points without external pre-training. This is achieved with a weight-tied attention cell (self-attention over property tokens plus cross-attention to chemistry embeddings) trained with per-cycle supervision; physics-informed sensors provide supporting priors, and gate-based halting supplies a zero-parameter convergence signal.

Across six Matbench tasks, TRIADS achieves 91.20 ± 12.23 **MPa** on `matbench_steels`. It also reports 0.3068 ± 0.0082 **eV** on `matbench_expt_gap` and 35.89 ± 12.40 **meV/atom** on `matbench_jdft2d`. For classification, TRIADS reaches 0.9655 ± 0.0029 ROC-AUC on `matbench_expt_is_metal` and 0.9285 ± 0.0063 on `matbench_glass`; on `matbench_phonons`, it attains 41.91 ± 4.04 **cm⁻¹**. On matched benchmarks, TRIADS outperforms CrabNet, AMMExpress, RF-SCM, Magpie, GPTChem, and Darwin; on `matbench_jdft2d`, TRIADS also outperforms pre-trained coGN and coNGN while using no external pre-training.

Key controlled result on `matbench_steels`: with architecture and features fixed, adding per-cycle deep supervision to the same recursive attention model reduces MAE from 134.59 to 103.29 **MPa** (**23.3%** relative). This directly identifies recursive attention + deep supervision as the highest-leverage combination in our small-data setting.

Evaluation on the remaining Matbench benchmarks is deferred due to compute limits.

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1 Introduction

Core claim. In small-data materials ML, performance depends on the pairing of informative sensors and architecture: physics-grounded inputs provide signal, while a weight-tied attention-based recursive cell, stabilized by GRU-style gating and trained with per-cycle deep supervision, refines that signal efficiently. On `matbench_expt_is_metal`, TRIADS reaches ROC-AUC 0.9655 ± 0.0029 with 100K parameters, and the 44K variant still reaches 0.9644 (6.79 points above GPTChem at $\gg 1$ B parameters). A controlled ablation on `matbench_steels` shows that, with architecture and features fixed, per-cycle supervision reduces MAE from 134.59 to 103.29 MPa (23.3% relative).

The small-data setting in materials ML. Materials science benchmarks are structurally different from regimes where deep learning usually excels. While ImageNet has 1.2 million images and language corpora contain billions of tokens, canonical Matbench tasks [7] contain only 312 to 5,680 samples. This is structural, not temporary scarcity: each sample requires wet-lab synthesis or expensive DFT computation. When a training fold fits in a single GPU batch, larger models are often the wrong inductive bias.

Most work addresses this by pre-training large models on aggregated databases. Graph neural networks such as MEGNet [2] (~ 2 M parameters), ALIGNN [4] (~ 4 M parameters), and the coGN/coNGN family [12] (~ 1 M parameters) perform strongly on structural benchmarks partly because they are pre-trained on hundreds of thousands of DFT calculations. This strategy is valid, but it assumes a large relevant pre-training corpus and often sidesteps the small-data setting instead of directly addressing it.

The TRIADS approach. We take a different route: prioritize *architectural inductive bias* and training dynamics over capacity scaling. Our recursive architecture, a single shared cell iterated T times, extracts signal from limited data through four mechanisms:

- **Attention-based recursive cell:** Unlike MLP-style recurrent cells, each TRIADS cycle recomputes relational structure through self-attention over property/graph tokens plus cross-attention to chemistry context.
- **Weight-tied recurrence:** Shared parameters across cycles enforce a general-purpose update rule rather than cycle-specific transformations, providing strong regularization for small-data regimes.
- **GRU-style gated residuals:** Learned update gates stabilize deep-cycle optimization and prevent latent-state divergence.
- **Deep supervision:** Computing a weighted loss at every one of T cycles multiplies effective gradient updates by a factor of T , without adding training data. For $T = 20$ on a 200-sample fold, this is equivalent to providing 20 supervision signals per sample.
- **Implicit gate-based halting:** Existing gate activations are reused as a zero-parameter convergence signal, avoiding a separate halting network.

Task-specific sensors and architecture are complementary: sensors inject physically grounded priors, while the coupled recursive design determines how effectively those priors are converted into predictions. With identical features and model width on `matbench_steels`, this design yields a 23.3% MAE reduction from supervision alone ($134.59 \rightarrow 103.29$ MPa).

What is novel in the combination. Prior TRM-style recurrence emphasizes lightweight MLP update cells [13]. Prior work has applied graph networks to materials [2, 4, 12, 17]. Deep supervision has been used in other domains. TRIADS combines attention-based recursive cells, hierarchical graph recursion *inside* a weight-tied loop, and per-cycle deep supervision in a design motivated by the small-data constraint. To our knowledge, this specific combination has not been applied to materials property prediction, and the controlled evidence in Section 4.7 suggests it is the combination, not any single component, that drives the observed efficiency.

Architecture overview. TRIADS has two variants. **HybridTRIADS** (Section 3.5) handles composition-only tasks using Magpie/Mat2Vec tokens with self- and cross-attention. **GraphTRIADS** (Section 3.6) handles structural tasks using a 3-order hierarchical graph (atoms, bonds, angles, dihedrals) with physics-featurized edges inside the same recursive loop. The main contribution is the shared recursive backbone and its integration with modality-specific sensors, which act as physically informed input adapters.

Contributions.

1. **Unified architectural design for small-data materials ML:** We introduce (a) an attention-based recursive cell for compositional reasoning (self-attention over property tokens + cross-attention to chemistry embeddings), (b) a structural variant where hierarchical graph message passing (atoms $\rightarrow$ bonds $\rightarrow$ angles $\rightarrow$ dihedrals) is the weight-tied recursive cell itself, and (c) a coupled training design in which per-cycle deep supervision works with these cells to produce calibrated iterative predictions.
2. **Benchmark performance:** Across six Matbench tasks, TRIADS attains strong accuracy at 44K–247K parameters, including ROC-AUC 0.9655 ± 0.0029 on `matbench_expt_is_metal` and MAE 0.3068 ± 0.0082 eV on `matbench_expt_gap`; supporting results include MAE 35.89 ± 12.40 meV/atom on `matbench_jdft2d` (5-seed ensemble).
3. **Architecture-coupled training principle:** With identical architecture on `matbench_steels`, supervising every reasoning cycle reduces MAE from 134.59 to 103.29 MPa (23.3% relative), showing that deep supervision is a high-leverage component within this recursive design.
4. **Failure-mode analysis:** An HRM-inspired hierarchical variant collapses on `matbench_steels` (431.87 ± 49.59 MPa), showing that gradient-detached hierarchy is not robust in this small-data regime.

We release code, checkpoints, fold-level logs, and negative-result artifacts for reproducibility.

2 Background and Related Work

Matbench. Matbench [7] provides fixed 5-fold nested cross-validation splits for 13 materials property prediction tasks. We evaluate on six tasks (Table 1), following the exact protocol (random state 18012019).

Composition-based models. CrabNet [16] (~ 600 K parameters) treats elemental fractions as attention tokens and applies multi-head self-attention to model element interactions. Roost [9] models stoichiometric relations via graph attention over element nodes. MODNet [5] applies gradient-boosted trees over handcrafted matminer feature sets with feature selection; it has near-zero *neural* parameters but requires extensive per-dataset feature engineering.

Table 1: Six Matbench tasks. Most ML benchmarks operate at $\gg 100\text{K}$ samples; these are 312–5,680.

Task	Target	N	Type	Input modality
matbench_steels	Yield strength (MPa)	312	Reg.	Composition
matbench_expt_gap	Band gap (eV)	4,604	Reg.	Composition
matbench_expt_is_metal	Metal/non-metal	4,921	Cls.	Composition
matbench_glass	Glass forming ability	5,680	Cls.	Composition
matbench_jdft2d	Exfoliation energy (meV/atom)	636	Reg.	Comp. + struct.
matbench_phonons	Peak phonon freq. (cm^{-1})	1,265	Reg.	Structure

Structure-based GNNs. CGCNN [17] pioneered crystal graph convolution. MEGNet [2] ($\sim 2\text{M}$ params) added global state vectors updated jointly with atom and bond features. ALIGNN [4] ($\sim 4\text{M}$ params) introduced line-graph angular convolutions. coGN and coNGN [12] ($\sim 1\text{M}$ params) pushed state-of-the-art with connectivity-aware graph networks. All these models benefit substantially from pre-training on large DFT databases.

Adaptive and recursive computation. Universal Transformers [6] introduced weight-tied recurrence with learned halting. Adaptive Computation Time (ACT) [10] adds per-token halting probabilities through a ponder cost $\lambda\mathcal{P}$; in our small-data setting, this hyperparameter is highly sensitive ($\lambda = 0.01$ collapsed to immediate minimum-step halting). The Hierarchical Reasoning Model (HRM) [8] uses nested high- and low-level modules with gradient detachment; our adaptation to materials data fails catastrophically because gradients are blocked (Section 5.4). The Tiny Recursive Model (TRM) [13] offers a simpler single-network alternative. Here, “TRM” in HYBRIDTRIADS/GRAPHTRIADS refers to our materials-specific recursive module family, inspired by but not identical to [13]. TRIADS adopts this recurrence philosophy and adds task-specific featurization, gate-based halting, and systematic materials benchmarking.

3 The TRIADS Architecture

Design philosophy. Each TRIADS component is chosen for the small-data constraint, and the components are intentionally interdependent. We start with physics-informed sensors so the model receives task-relevant signal. We then use an attention-based shared cell (rather than an MLP-only cell) so each recursive update is relational and can re-interrogate interactions at every cycle. We tie weights across cycles (rather than stacking untied layers) to enforce one general-purpose update rule with strong regularization. We supervise every cycle (rather than only the final output) so each iteration must remain calibrated. For structural tasks, we place hierarchical graph message passing inside the loop (rather than as a pre- or post-hoc feature stage), making iterative geometric reasoning the cell itself.

3.1 Recursive Reasoning Core

The shared skeleton of both TRIADS variants is a recursive loop over a weight-tied cell f_θ :

$$\Delta_t = f_\theta(\mathbf{z}_{t-1}, \mathbf{x}) \quad (1)$$

$$\mathbf{g}_t = \sigma(W_g[\mathbf{z}_{t-1}; \Delta_t]) \quad (2)$$

$$\mathbf{z}_t = \mathbf{z}_{t-1} + \mathbf{g}_t \odot \Delta_t \quad (3)$$

where $\mathbf{x}$ is the input (feature vector or graph), $\mathbf{z}_0 = W_{\text{in}}\mathbf{x}$ is an initial linear projection, $W_g \in \mathbb{R}^{d \times 2d}$ is the gating projection, σ is the sigmoid function, and $\odot$ denotes elementwise multiplication. The prediction at cycle t is $\hat{y}_t = h(\mathbf{z}_t)$ where h is a lightweight head ($\mathbb{R}^d \rightarrow \mathbb{R}^c$, $c = 1$ for regression).

Weight sharing as regularization. Sharing parameters across cycles forces f_θ to learn one general update rule instead of T cycle-specific transformations. This deliberate capacity constraint strongly regularizes training. A stack of T untied layers with the same width has T times more parameters and is more prone to overfitting on our 200–1,000 training samples per fold. A T -cycle TRIADS model gives $T \times$ computational depth at the parameter cost of one cell. This trade-off is favorable in the small-data regime; on much larger datasets, untied deep stacks may be preferable.

3.2 Gated Residuals: Stabilizer and Halting Signal

Without gating, Eq. (3) reduces to $\mathbf{z}_t = \mathbf{z}_{t-1} + \Delta_t$. In all our experiments, plain additive residuals cause exponential latent state growth beyond 6–8 cycles, producing NaN gradients. With shared weights, gradients accumulate through T identical residual paths; the effective gradient magnitude scales as $O(T)$, causing instability for $T > 8$ without gating.

The sigmoid gate Eq. (2) is GRU-style [3]: it learns to throttle the update Δ_t based on whether the current state $\mathbf{z}_{t-1}$ warrants further refinement. When the state has converged, the gate closes ($\mathbf{g}_t \approx \mathbf{0}$). When substantial refinement remains, the gate opens ($\mathbf{g}_t \approx \mathbf{1}$). This gating is architecturally *necessary* for stable training beyond 8 cycles, not an optional enhancement. It is also a prerequisite for deep supervision at $T = 20$: without stable bounded updates, per-cycle losses amplify instability instead of useful gradient signal.

3.3 Implicit Gate-Based Halting

Gate activations $\mathbf{g}_t$ already encode convergence. When $\max_i g_t^{(i)} < \epsilon$, no state dimension wants to update, so computation has reached equilibrium. We halt at cycle t^* :

$$t^* = \min\{t \geq T_{\min} \mid \max_i g_t^{(i)} < \epsilon\}, \quad T_{\min} = 4, \epsilon = 0.05 \quad (4)$$

To encourage earlier convergence without destabilizing training, we add a gate sparsity term to the total loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{pred}} + \mathcal{L}_{\text{gate}}, \quad \mathcal{L}_{\text{gate}} = \gamma \cdot \frac{1}{T} \sum_{t=1}^T \bar{g}_t, \quad \gamma = 0.001 \quad (5)$$

where $\bar{g}_t$ is the mean gate activation at cycle t . This adds zero parameters beyond W_g already required for stability.

Comparison to ACT. ACT [10] requires a dedicated halting network ($\sim 2\text{K}$ additional parameters) and a ponder-cost hyperparameter λ . In our V11 experiments on `matbench.steels`, ACT with $\lambda = 0.01$ collapsed to a trivial solution: the ponder cost dominated the loss, teaching the model to halt as early as possible (mean halt step 15.6, 97% early halting), independently of sample difficulty. The result was MAE 132.59 MPa; negligibly better than no halting at all (134.59 MPa). Our gate-based mechanism avoids this because γ controls the *mean gate*

magnitude rather than the *stopping time*, making the regularizer less directly entangled with prediction accuracy. In all reported runs, we use $\gamma = 0.001$.

The main benefit of gate-based halting is simplicity and zero extra parameters, not a large accuracy gain. The 1.38 cm^{-1} difference between fixed and adaptive cycles on `matbench_phonons` (41.91 vs. 43.29) may be noise; we treat halting as an engineering feature (30–50% inference speedup) rather than an accuracy contributor.

3.4 Deep Supervision

Training minimizes a weighted sum of per-cycle predictions:

$$\mathcal{L}_{\text{pred}} = \sum_{t=1}^T w_t \cdot \ell(\hat{y}_t, y), \quad w_t = \frac{2t}{T(T+1)}, \quad \sum_t w_t = 1 \quad (6)$$

where ℓ is MAE for regression and binary cross-entropy for classification. The closed-form denominator $T(T+1)/2$ normalizes the weights. Linear ramping ($w_t \propto t$) gives later cycles the highest influence while still requiring useful intermediate predictions.

Architecture coupling. Deep supervision is especially effective here because f_θ is attention-based. Each cycle recomputes relational structure (self-attention plus cross-attention), so intermediate states are prediction-relevant rather than transient hidden activations. Supervising $\hat{y}_t$ therefore calibrates useful partial solutions cycle-by-cycle. We do not treat deep supervision as a generic add-on; in TRIADS it works as part of the coupled recursive design.

Why this matters for small datasets. With $N \approx 200$ training samples per fold on `matbench_steels` (our smallest task), final-step-only training provides one gradient signal per sample per epoch. Supervising all $T = 20$ cycles provides 20 signals per sample, effectively multiplying supervision density by T . This is analogous to multi-task learning without extra labels. Section 4.7 isolates this effect: with architecture fixed, deep supervision reduces MAE by 23.3% ($134.59 \rightarrow 103.29 \text{ MPa}$).

We hypothesize this gain would be weaker for less relational recursive cells (e.g., plain MLP-only updates), because their intermediate states are less likely to be calibrated for prediction at each step; testing this directly is left to future cell-level ablations.

Deep supervision also prevents over-refinement. Without it, a 20-cycle model trained only on the final prediction can over-fit by cycles 16–20 while intermediate predictions receive no gradient. With deep supervision, every cycle must remain calibrated, preventing runaway refinement.

3.5 HybridTRIADS: The Compositional Variant

HYBRIDTRIADS handles properties governed by elemental composition. No crystal structure is required as input. The model operates by tokenizing elemental properties into an attention-compatible representation, reasoning over inter-property interactions through self-attention, and injecting chemistry context through cross-attention.

3.5.1 Input Representation

Each composition $\mathbf{c}$ (a stoichiometric formula) is featurized into a fixed-dimension vector:

$$\mathbf{x}_{\text{comp}} = \left[\underbrace{\mathbf{x}_{\text{Magpie}}}_{132\text{d}} \parallel \underbrace{\mathbf{x}_{\text{Mat2Vec}}}_{200\text{d}} \parallel \underbrace{\mathbf{x}_{\text{extras}}}_{22\text{d}} \right] \in \mathbb{R}^{354} \quad (7)$$

Magpie descriptors [15]: 132 composition statistics (22 elemental properties, each summarized by mean/std/min/max/range/mode).

Mat2Vec embeddings [14]: a 200-d composition vector from weighted element embeddings, $\mathbf{x}_{\text{Mat2Vec}} = \sum_i c_i \mathbf{e}_i$.

Physics extras (22d): stoichiometry entropy, valence-orbital and ionic statistics, and transition-metal fraction. For electronic tasks (`matbench_expt_gap`, `matbench_expt_is_metal`), we additionally include BandCenter and HOMO/LUMO-gap proxies; for `matbench_glass`, these electronic extras are excluded (≈ 351 features total).

3.5.2 Architecture of the Shared Cell f_θ

The cell f_θ in HYBRIDTRIADS maps $(\mathbf{z}_{t-1}, \mathbf{x})$ to Δ_t through four sequential operations. Unlike MLP-centered TRM cells, this shared cell is attention-native: every cycle explicitly re-computes inter-property relations and chemistry-conditioned context.

(1) Magpie tokenization. The 132-dim Magpie vector is reshaped into 22 property tokens $\{\mathbf{p}_j\}_{j=1}^{22} \in \mathbb{R}^6$, where property j has token $[p_{j,\text{mean}}, p_{j,\text{std}}, p_{j,\text{min}}, p_{j,\text{max}}, p_{j,\text{range}}, p_{j,\text{mode}}]$. These are linearly projected to the model dimension d :

$$\mathbf{h}_j^{(0)} = W_{\text{tok}} \mathbf{p}_j + \mathbf{b}_{\text{tok}} \in \mathbb{R}^d, \quad j = 1, \dots, 22 \quad (8)$$

(2) Self-attention over property tokens (1–2 layers). Multi-head self-attention (MHA) over the 22 property tokens captures cross-property interactions, e.g., how the standard deviation of electronegativity correlates with the mean valence electron count:

$$\mathbf{H}^{(1)} = \text{LayerNorm}\left(\mathbf{H}^{(0)} + \text{MHA}(\mathbf{H}^{(0)})\right) \quad (9)$$

$$\mathbf{H}^{(1)} = \text{LayerNorm}\left(\mathbf{H}^{(1)} + \text{FFN}(\mathbf{H}^{(1)})\right) \quad (10)$$

where the FFN is a 2-layer MLP with GELU activations and hidden dim $d_{\text{ff}} = 4d$. When 2 SA layers are used (V13A, the best steels result), the second layer captures 2nd-order property interactions. The aggregate representation is mean-pooled: $\mathbf{s} = \frac{1}{22} \sum_j \mathbf{H}_j^{(1)} \in \mathbb{R}^d$.

(3) Cross-attention with Mat2Vec chemistry embedding. The current latent state $\mathbf{z}_{t-1}$ queries the Mat2Vec composition embedding to inject chemical context:

$$\mathbf{q} = W_Q \mathbf{z}_{t-1}, \quad \mathbf{k} = W_K \mathbf{x}_{\text{Mat2Vec}}, \quad \mathbf{v} = W_V \mathbf{x}_{\text{Mat2Vec}} \quad (11)$$

$$\boldsymbol{\alpha} = \text{softmax}\left(\mathbf{q}\mathbf{k}^\top / \sqrt{d}\right), \quad \mathbf{c}_t = \boldsymbol{\alpha}\mathbf{v} \quad (12)$$

The output fuses the reasoned property representation with a chemistry-aware context: $\Delta_t = W_{\text{out}}[\mathbf{s}; \mathbf{c}_t]$.

(4) MLP reasoning block. Before the gated update, a feedforward block applies a final nonlinear transformation: $\Delta_t \leftarrow W_2 \text{GELU}(W_1 \Delta_t + b_1) + b_2$.

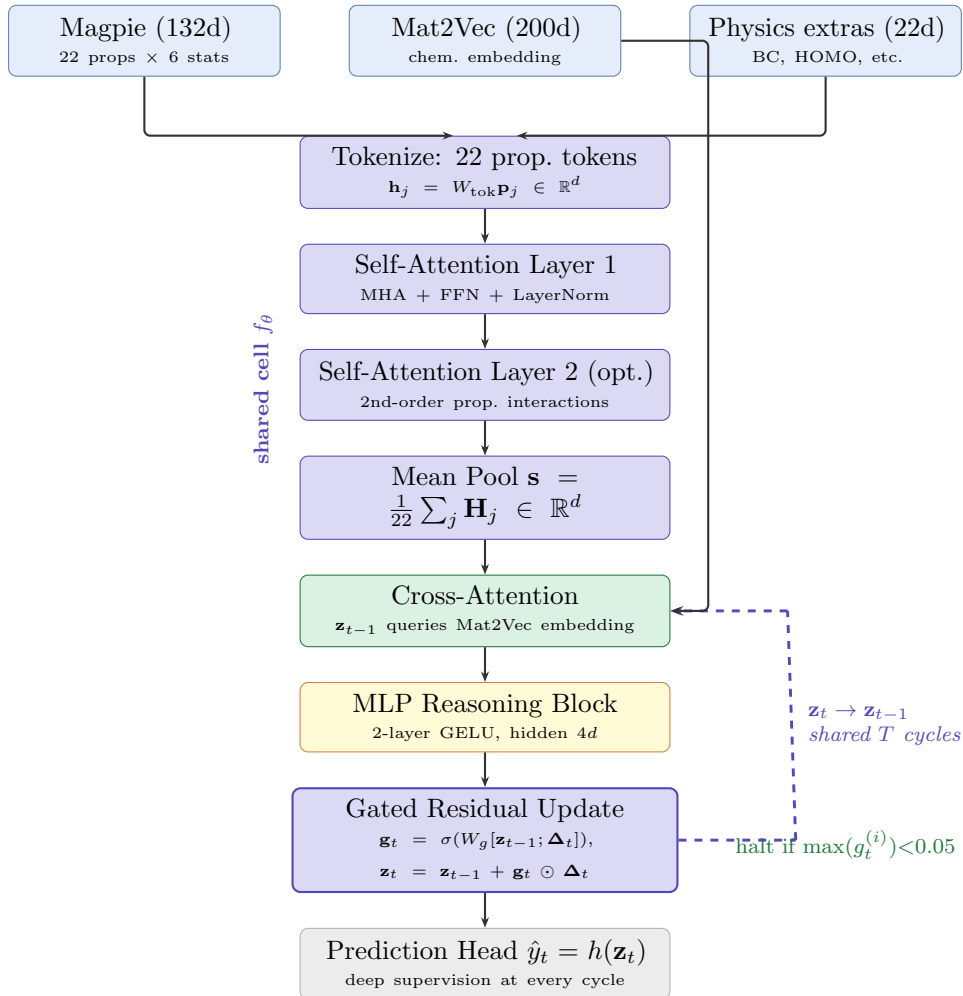


Figure 1: HYBRIDTRIADS architecture. The shared cell f_θ is cycled T times (16–20 cycles). At each cycle, self-attention over 22 Magpie property tokens captures inter-property correlations; cross-attention injects Mat2Vec chemistry context; the gated residual update (bold box) stabilizes training and provides a halting signal. Deep supervision computes a loss at every prediction head output $\hat{y}_t$.

Gated update and prediction. Eqs. (2)–(3) are applied to produce $\mathbf{z}_t$; then $\hat{y}_t = W_{\text{head}} \text{ReLU}(W_h \mathbf{z}_t)$.

Figure 1 shows the complete HYBRIDTRIADS cell and recurrence loop used across composition tasks.

Training configuration. AdamW, lr 5×10^{-4} , weight decay 10^{-4} , cosine annealing with 10-epoch linear warmup, 200–300 epochs, batch size 32–64, dropout 0.10–0.20. SWA [11] from epoch 150 onward. 5-fold CV per Matbench protocol. Model dimensions: $d_{\text{attn}} \in \{24, 32, 40, 64\}$, $d_{\text{hidden}} \in \{48, 64, 72\}$, 4 attention heads. Parameter counts: 44K–225K.

3.6 GraphTRIADS: The Structural Variant

GRAPHTRIADS handles vibrational properties requiring 3D structural information. It wraps a 3-order hierarchical graph neural network inside the recursive loop. The hierarchical graph

message-passing stack is the shared recurrent cell itself: the same atom→bonds→angles→dihedrals operators are re-applied with tied weights at each cycle.

3.6.1 Why Geometry Is Necessary for Phonons

The highest optical phonon frequency $\omega_{\max}$ is set by the stiffest bond mode in the crystal. In a classical harmonic approximation, $\omega_{\max} \propto \sqrt{k_{\max}/\mu_{\min}}$ where k is a force constant and μ is a reduced mass. A composition-only model receives elemental masses m_i but cannot observe inter-atomic force constants, which depend critically on bond lengths, bond angles, and coordination geometry. This is not a modeling limitation; it is a physical one. No amount of model capacity can recover missing geometric information.

Our V1 bag-of-atoms TRM (71.82 cm⁻¹) confirms this: despite extracting composition-level physics, performance remained substantially below top GNN baselines. The subsequent ablation (Table 9) shows that each geometric sensor adds predictable, physically interpretable information.

3.6.2 3-Order Hierarchical Graph

We model each crystal as a layered multi-body graph at three orders:

- **Order 1 (Atom graph $\mathcal{G}_1$):** Nodes are atoms; edges connect the 12 nearest neighbors within 8 Å. Node features: element physics (mass m , χ , radius r , IE_1 , EA , valence, group, period, block, is_metal) + fractional coordinates + coordination statistics = 18 features.
- **Order 2 (Bond/Line graph $\mathcal{G}_2$):** Nodes are directed bonds ($i \rightarrow j$); edges connect bonds ($i \rightarrow j$) and ($j \rightarrow k$) that share atom j , encoding triplet angles $\theta_{ijk} = \angle(\mathbf{r}_i - \mathbf{r}_j, \mathbf{r}_k - \mathbf{r}_j)$.
- **Order 3 (Dihedral graph $\mathcal{G}_3$):** Nodes are bond triplets (i, j, k); edges connect triplet pairs sharing a bond, encoding dihedral torsion angles ϕ_{ijkl} .

3.6.3 Physics-Informed Edge Features

For each bond (i, j) we pre-compute empirical proxy features:

$$k_{ij} \approx \bar{\chi}_{ij}/r_{ij}^2, \quad \mu_{ij} = m_i m_j / (m_i + m_j), \quad \omega_E^{(ij)} = \sqrt{k_{ij}/\mu_{ij}} \quad (13)$$

Caveat on k_{ij} : $k \sim \bar{\chi}/r^2$ is an empirical proxy, not a standard force constant formula. The standard Badger’s rule [1] gives $k \propto r^{-3}$; other variants use atomic charge. Our formula is motivated by the general scaling of bond stiffness with electronegativity and inversely with bond length squared; it is validated empirically by the improvement it provides (56.33 → 41.91 cm⁻¹ from V4 to V6). The Einstein frequency estimate $\omega_E^{(ij)} = \sqrt{k_{ij}/\mu_{ij}}$ gives a classical per-bond zero-order phonon frequency; the neural network then learns quantum corrections.

Complete edge feature dimensions:

- Bond (Order 1 → 2): $[k_{ij}, \mu_{ij}, \omega_E^{(ij)}, \Delta\chi_{ij}, \text{ionicity}, r_i/r_j, m_i/m_j, 1/r_{ij}]$ (8d) + Gaussian RBF over r_{ij} (40d) + unit direction $\hat{\mathbf{r}}_{ij}$ (3d) = 51d
- Angle (Order 2 → 3): Gaussian RBF over θ_{ijk} in $[0, \pi]$ (8d)
- Dihedral (Order 3): Gaussian RBF over ϕ_{ijkl} in $[0, \pi]$ (8d)
- Global context (15d): Debye temperature estimate $\theta_D \approx \hbar\omega_E/k_B$, mean and std of k_{ij} , mass variance $\text{Var}(m)$, electronegativity variance, mean coordination number, density, volume/atom, packing fraction, mean/std bond length, max/min atom mass.

- Composition context (361d): 132d Magpie + extras + 11d structural scalars (space group, lattice parameter statistics, symmetry class) + 200d Mat2Vec.

3.6.4 Recursive GraphTRIADS Loop

Algorithm 1 gives the full forward pass.

Algorithm 1 GRAPHTRIADS Forward Pass

Require: Crystal graph $(\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3)$, composition features $\mathbf{x}_c$, max cycles T

Ensure: Prediction $\hat{y}$

```

1: Init: Project atom features  $\mathbf{h}_i^{(0)} \in \mathbb{R}^d$ , bond features  $\mathbf{e}_{ij}^{(0)} \in \mathbb{R}^d$ , global  $\mathbf{z}_0 = W_{\text{in}}\mathbf{x}_c$ 
2: Warm-up: Apply 1 unshared GNN layer to  $\mathcal{G}_1$  (independent weights, structural bootstrap)
3: for  $t = 1, 2, \dots, T$  do ▷ Shared weights across all cycles
4:   P1: Dihedral  $\rightarrow$  Angle  $\mathbf{e}^{\text{ang}} \leftarrow \text{MLP}(\mathbf{e}^{\text{ang}} + \text{Agg}_{\mathcal{G}_3}(\mathbf{e}^{\text{dih}}))$ 
5:   P2: Angle  $\rightarrow$  Bond  $\mathbf{e}_{ij} \leftarrow \text{MLP}(\mathbf{e}_{ij} + \text{Agg}_{\mathcal{G}_2}(\mathbf{e}^{\text{ang}}))$  ▷ Bond embeddings now carry angular stiffness
6:   P3: Bond  $\rightarrow$  Atom (edge-gated conv)
7:      $\mathbf{m}_i \leftarrow \sum_{j \in \mathcal{N}(i)} \sigma(W_e \mathbf{e}_{ij}) \odot W_v \mathbf{h}_j$  ▷ Bond embedding gates atom message
8:      $\mathbf{h}_i \leftarrow \text{LayerNorm}(\mathbf{h}_i + \text{MLP}(\mathbf{m}_i))$ 
9:   P4: Joint Self-Attention over  $[\mathbf{h}_1, \dots, \mathbf{h}_n, \mathbf{z}_{t-1}]$  ▷ Global token aggregates long-range cross-site context
10:  P5: Cross-Attention Comp. tokens (Q)  $\leftarrow$  Atoms (K, V) ▷ Chemistry grounds local geometry
11:   $\Delta_t \leftarrow W_{\text{agg}}[\text{ReadOut}(\mathbf{h}); \mathbf{z}_{t-1}; \mathbf{x}_c^{\text{global}}]$ 
12:  P6: Gated Residual Update  $\mathbf{g}_t = \sigma(W_g[\mathbf{z}_{t-1}; \Delta_t])$ ,  $\mathbf{z}_t = \mathbf{z}_{t-1} + \mathbf{g}_t \odot \Delta_t$ 
13:  P7: Prediction  $\hat{y}_t = h(\mathbf{z}_t)$  ▷ Deep supervision: contribute to loss
14:  if  $t \geq T_{\text{min}}$  and  $\max_i g_t^{(i)} < 0.05$  then break ▷ Gate-based halt
15:  end if
16: end for
17: return  $\hat{y}_t$ 

```

Training. AdamW, lr 5×10^{-4} , weight decay 10^{-4} , cosine annealing (10-epoch warmup), 200 epochs, batch 64, dropout 0.10, gradient clip 0.5, SWA from epoch 150. $D = 64$, 4 heads, 8 fixed cycles, 1 unshared warm-up layer. Total: 247K parameters.

4 Experiments

All TRIADS parameter counts are for the neural model only. “tree-based” denotes gradient-boosted tree ensembles applied to hand-crafted matminer features; their neural parameter count is essentially zero (the complexity lies in feature engineering, not neural weights).¹

4.1 Matbench Steels: Yield Strength

matbench_steels (312 samples, ≈ 200 per training fold) is the most data-scarce regression task.

TRIADS (91.20 ± 12.23 MPa) outperforms RF-SCM/Magpie (103.51), CrabNet (107.32), and Darwin (123.29), and trails MODNet v0.1.12 (87.76) by 3.44 MPa. MODNet uses exhaustive feature selection and gradient-boosted trees without per-sample neural computation; TRIADS uses one fixed neural architecture across benchmarks.

The high fold standard deviation (± 12.23 MPa) reflects real data heterogeneity: fold 1 contains harder alloys (MAE 114.32), while folds 2–3 are easier (80–81 MPa). This pattern

¹MODNet specifically uses feature selection over matminer descriptors and gradient-boosted trees with near-zero neural parameters. Comparisons on small datasets should account for the fact that tree ensembles can be very strong when the right feature set is identified.

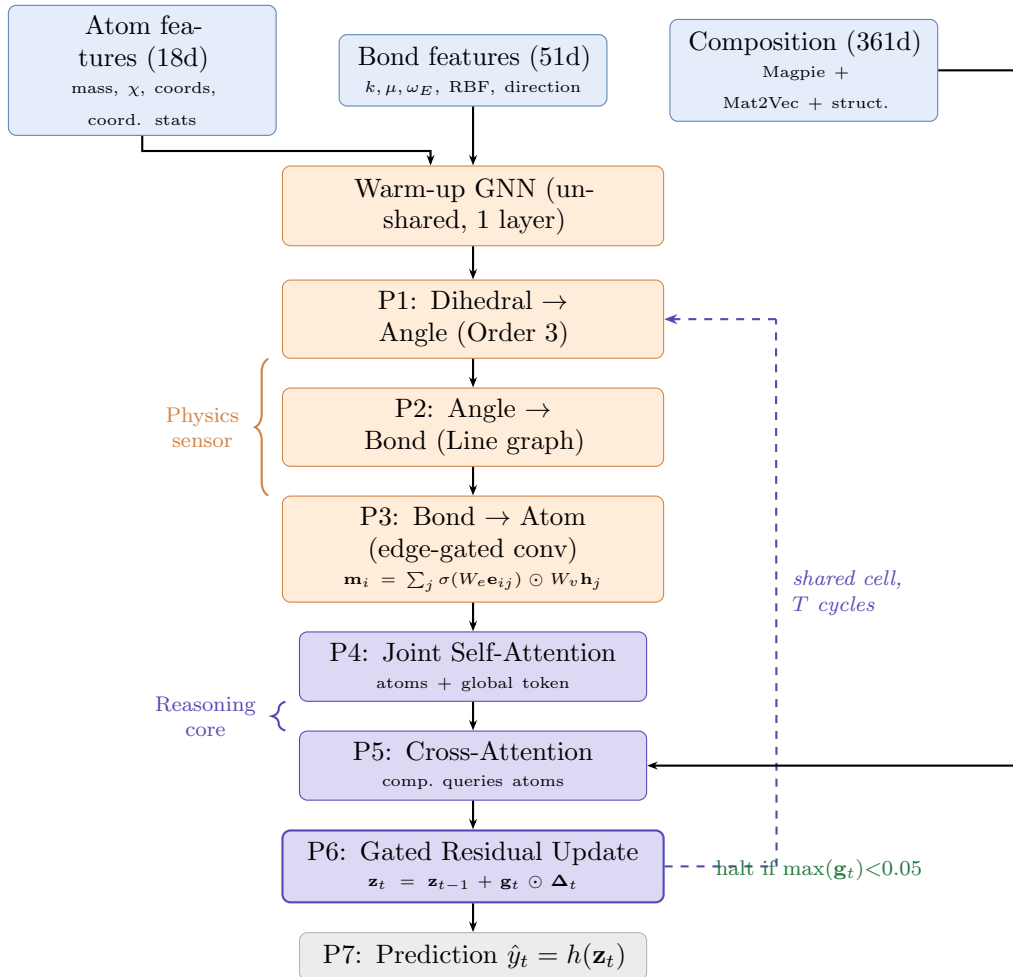


Figure 2: GRAPHTRIADS recursive loop. Phases P1–P3 (orange) process the 3-order hierarchical graph, propagating physical information from dihedrals to bonds to atoms. Phases P4–P5 (purple) apply attention over the resulting atom embeddings. Phase P6 applies the gated residual update, which also provides the halting signal. All phases within the loop share weights across $T = 8$ fixed cycles (or 4–12 adaptive).

appears across TRIADS versions and baselines. Seed variance is low: a 3-seed check gives 105.85 ± 3.00 MPa, indicating stable initialization behavior.

Table 3 traces the full version history, showing the contribution of each architectural improvement.

Key observations. Table 3 shows a two-stage effect. First, introducing the attention-based hybrid cell (V3/V5A $\rightarrow$ V7B) gives a modest MAE reduction ($130.33/128.98 \rightarrow 127.08$), but changes the recursive update dynamics. Second, on that cell, per-cycle deep supervision produces the dominant jump (V9A $\rightarrow$ V10A: $134.59 \rightarrow 103.29$ MPa) with identical width and features. In this trajectory, the attention-based cell is the prerequisite and deep supervision is the unlock.

Without deep supervision, raising T from 16 to 20 *hurts* performance (V7B $127.08 \rightarrow$ V9A 134.59) by over-refining easy folds (fold 3: $104.59 \rightarrow 142.55$; fold 5: $109.78 \rightarrow 131.31$). Deep supervision removes this failure mode.

Table 2: `matbench_steels`: MAE (MPa), lower is better. TRIADS V13A fold-level: [114.32, 81.46, 80.55, 90.49, 89.18] (5-seed ensemble per fold). Dummy predictor: 229.74 MPa.

Model	Method	MAE (MPa)	Params
Dummy (mean)	–	229.74	0
TPOT-Mat	AutoML	79.95	–
AutoML-Mat	AutoML	82.30	–
MODNet v0.1.12	tree-based	87.76	~0
RF-Regex Steels	RF	90.59	–
TRIADS V13A[†] (5-seed)	Recursive attn.	91.20 ± 12.23	225K
MODNet v0.1.10	tree-based	96.21	~0
AMMExpress v2020	RF/AutoML	97.49	–
RF-SCM/Magpie	RF	103.51	–
CrabNet [†]	Transformer	107.32	~600K
Darwin	Evolutionary	123.29	–

[†] Composition-only neural models. TRIADS is a 5-seed fold-wise ensemble; CrabNet is the single published leaderboard entry.

Table 3: TRIADS steels development: each version adds exactly one improvement over the prior best. All use the same Matbench 5-fold protocol.

Version	Key change	MAE (MPa)	Params
V1 (Mat2Vec MLP)	Baseline, mat2vec only input	184.38	100K
V3 (Magpie MLP)	Magpie features replace mat2vec	130.33	248K
V5A (+ SWA)	Stochastic weight averaging	128.98	67K ^a
V7B (Hybrid-L)	Self-attn + cross-attn cell, $T = 16$	127.08	87K
V9A (same, $T = 20$)	Extend to 20 cycles (no DS)	134.59	87K
V10A (+ DS, $T = 20$)	Deep supervision at every cycle	103.29	87K
V11B (scaled)	Wider model ($d = 64$, 172K params)	102.30	172K
V12A (+ features)	Expanded feature set, 191K	95.99	191K
V13A (+ 2-SA, ens.)	2nd SA layer, 5-seed ensemble	91.20	225K

^aV5A used a compact MLP-SWA (67K) to test SWA in isolation; the 87K Hybrid cell introduced in V7B is a different, larger architecture with SA+CA layers.

4.2 `matbench_expt_gap`: Experimental Band Gap

TRIADS (0.3068 ± 0.0082 eV) outperforms CrabNet (0.3463), Ax/SAASBO CrabNet (0.3310), and MODNet v0.1.12 (0.3327). Among confirmed composition-only models in Table 4, TRIADS is strongest. Darwin’s public entry (0.2865) is shown separately because its input modality is undocumented.

Design lesson: replacing a generic 468-feature input with a 354-feature physics-informed set (adding BandCenter and HOMO/LUMO gap, removing ElementFraction) improves single-seed MAE from 0.3616 to 0.3342 (7.6% relative) with *no architectural change*. Five-seed ensembling further reduces MAE to 0.3068 (additional 7.9%).

4.3 `matbench_expt_is_metal`: Metallicity Classification

TRIADS 100K achieves ROC-AUC 0.9655 ± 0.0029 . In the March 2026 leaderboard snapshot, this is 4.46 points above AMMExpress v2020 (0.9209). The 44K model (0.9644) is effectively

Table 4: `matbench_expt_gap`: MAE (eV), lower is better. TRIADS fold-level: [0.3069, 0.3122, 0.3190, 0.2998, 0.2962]. †: confirmed composition-only. ‡: modality unconfirmed.

Model	Method	MAE (eV)	Params
<i>Confirmed composition-only models</i>			
Ax/SAASBO CrabNet [†]	BO-tuned Transformer	0.3310	~600K
MODNet v0.1.12	tree-based	0.3327	~0
TRIADS V3 100K (5-seed)[†]	Recursive attn.	0.3068 ± 0.0082	100K
CrabNet [†]	Transformer	0.3463	~600K
AMMExpress v2020	RF/AutoML	0.4161	–
<i>Unconfirmed input modality (listed for completeness)</i>			
Darwin [‡]	Evolutionary	0.2865	–

[†] Composition-only entries. [‡] Darwin public entry does not explicitly confirm whether structure features are used for this benchmark.

Table 5: `matbench_expt_is_metal`: ROC-AUC, higher is better. TRIADS 100K fold-level: [0.9674, 0.9648, 0.9602, 0.9683, 0.9669]. †: composition-only.

Model	Method	ROC-AUC	Params
TRIADS 100K[†] (1 seed)	Recursive attn.	0.9655 ± 0.0029	100K
TRIADS 44K[†] (1 seed)	Recursive attn.	0.9644	44K
Darwin	Evolutionary	0.9598	–
AMMExpress v2020	RF/AutoML	0.9209	–
RF-SCM/Magpie	RF	0.9167	–
MODNet v0.1.12	tree-based	0.9161	~0
GPTChem	LLM prompting	0.8965	≫1B

identical within noise, showing that 44,000 parameters can suffice when featurization encodes relevant physics. Fold variance (± 0.0029) is low; additional seeds were omitted for compute reasons.

The 44K vs 100K comparison most clearly supports the sensor hypothesis: BandCenter and HOMO/LUMO gap features provide direct electronic-structure priors, so extra parameters add little.

4.4 Matbench Glass: Glass Formation Classification

TRIADS 44K (0.9285 ± 0.0063) outperforms AMMExpress (0.8607, +6.78pp) and RF-SCM/Magpie (0.8587, +6.98pp), and trails MODNet v0.1.12 (0.9603) by 3.18 points. MODNet’s edge likely reflects exhaustive feature selection tuned for this noisy thermodynamic classification target.

That 44K (0.9285) outperforms 100K (0.9259) is consistent with noise-limited targets: glass formation depends on undercooled-liquid thermodynamics and kinetics not captured in this dataset. Smaller models generalize better in this regime. The glass featurizer also excludes BandCenter and HOMO/LUMO gap features used for metallicity because mixing thermodynamics, not electronic structure, is the relevant prior.

Table 6: `matbench_glass`: ROC-AUC, higher is better. TRIADS 44K fold-level: [0.9244, 0.9222, 0.9245, 0.9392, 0.9322]. †: composition-only.

Model	Method	ROC-AUC	Params
MODNet v0.1.12	tree-based	0.9603	~0
TRIADS 44K (5-seed)[†]	Recursive attn.	0.9285 ± 0.0063	44K
TRIADS 100K (5-seed)[†]	Recursive attn.	0.9259 ± 0.0091	100K
AMMExpress v2020	RF/AutoML	0.8607	–
RF-SCM/Magpie	RF	0.8587	–
MODNet v0.1.10	tree-based	0.8107	~0

Table 7: `matbench_jdft2d`: MAE (meV/atom), lower is better. TRIADS V4 is a 5-seed fold-wise ensemble of the same 75K model as V3 (seed 42). †: composition + scalar structural features. ††: no pre-training; uses only in-fold Matbench data.

Model	Method	MAE	Params
<i>Tree-based models</i>			
MODNet v0.1.12	tree-based	33.19	~0
MODNet v0.1.10	tree-based	34.54	~0
<i>Pre-trained GNNs (not directly comparable)</i>			
coNGN	Pre-trained GNN	36.17	~1M
coGN	Pre-trained GNN	37.17	~1M
ALIGNN	Pre-trained GNN	43.42	~4M
<i>No pre-training^{††}</i>			
TRIADS V4[†] (5-seed ensemble)	Recursive attn.	35.89 ± 12.40	75K
TRIADS V3[†] (seed 42)	Recursive attn.	37.00 ± 11.11	75K
AMMExpress v2020	RF/AutoML	39.85	–
CrabNet	Transformer	45.61	~600K

4.5 Matbench JDFT2D: Exfoliation Energy

TRIADS V4 ensemble (35.89 ± 12.40 meV/atom) improves over V3 (37.00 ± 11.11) and outperforms coGN (37.17) and coNGN (36.17) in mean MAE without pre-training. Across seeds, run-level means are tightly clustered (37.92 ± 1.29), while fold 2 remains high for every seed (54–63 meV/atom), indicating a hard structural split rather than optimizer instability. The model uses Magpie/Mat2Vec composition features plus 11 scalar structural descriptors, not a full crystal graph.

4.6 Matbench Phonons: Peak Phonon Frequency

4.6.1 Sensor Ablation: Controlled Physics Validation

Table 9 is a controlled ablation: each version changes one sensor class while architecture (recursive cell, gating, deep supervision), optimizer, and hyperparameters stay fixed.

Each transition is physically interpretable:

V1 $\rightarrow$ V3.1 ($\Delta = -8.82 \text{ cm}^{-1}$): Adding bond distances provides atoms with positional context. The residual limitation: distance-only GNNs cannot distinguish tetrahedral ($\angle = 109.5$) from square-planar ($\angle = 90$) coordination: same nearest-neighbor distances, different angular

Table 8: JDFT2D fold-level results for TRIADS 75K. V4 ensemble improves V3 single-seed MAE by 5.4% (37.00 $\rightarrow$ 35.89). Fold 2 is consistently difficult across seeds.

Fold	Seed 42	Seed 123	Seed 456	Seed 789	Seed 1024	Ensemble
0	25.75	20.92	25.57	20.77	34.48	23.40
1	35.58	39.67	34.03	37.89	37.46	34.09
2	54.63	60.00	54.69	62.98	59.46	56.63
3	25.42	25.96	24.90	26.09	24.71	23.75
4	43.63	41.40	44.37	41.89	45.65	41.57
Mean $\pm$ Std	37.00 $\pm$ 11.11	37.59 $\pm$ 13.67	36.71 $\pm$ 11.43	37.92 $\pm$ 14.68	40.35 $\pm$ 11.67	35.89 $\pm$ 12.40

Table 9: Phonon sensor ablation. One sensor class is added per version; everything else is identical.

Version	Sensor added	MAE (cm^{-1})	Params
V1: Bag-of-Atoms	Composition tokens only	71.82	75K
V3.1: Distance GNN	Bond distances (RBF, CGCNN-style)	63.00	184K
V4: Angular GNN	Triplet angles θ_{ijk}	56.33	212K
GRAPHTRIADS V6 (fixed)	k, μ, ω_E ; dihedrals	41.91	247K

stiffnesses.

V3.1 $\rightarrow$ V4 ($\Delta = -6.67 \text{ cm}^{-1}$): Adding explicit triplet angles θ_{ijk} adds angular stiffness information that governs transverse and shear modes. This is the largest single-sensor gain, consistent with angular kinematics as the dominant missing variable for ω_{max} .

V4 $\rightarrow$ V6 ($\Delta = -14.42 \text{ cm}^{-1}$): Adding pre-computed Einstein frequencies $\omega_E^{(ij)}$, dihedral angles, and the full recursive graph-TRM framework. This step involves multiple concurrent improvements and its individual contributions cannot be cleanly separated.

If prediction quality scales with sensor quality, MAE should decrease monotonically; it does: $71.82 > 63.00 > 56.33 > 41.91$.

4.6.2 Full Leaderboard Comparison

GRAPHTRIADS V6 fixed-cycle ($41.91 \pm 4.04 \text{ cm}^{-1}$) outperforms gate-halt (43.29 ± 4.82) on 4 of 5 folds, with fold 1 as the exception. Fold 0 (45.06) and fold 4 (47.55) are hardest. Against non-pre-trained references, fixed-cycle V6 outperforms CrabNet (55.11), AMMExpress (56.17), and RF-SCM/Magpie (67.61), and approaches MODNet v0.1.10 (38.75). The remaining gap to pre-trained models (e.g., MEGNet 28.76) matches data scale: MEGNet is pre-trained on about 130K Materials Project structures, while each `matbench_phonons` fold gives GRAPHTRIADS about 1,012 structures without external pre-training (about $130\times$ fewer).

Figure 3 plots the progression with reference lines.

4.7 Deep Supervision Ablation

Table 12 isolates deep supervision on `matbench_steels` (V9/V10 experiments). Both rows use the *identical* architecture (87K parameters, 20 cycles); only the training loss changes.

Deep supervision reduces MAE by 23.3% with no architectural change. Per-fold results reveal the mechanism: folds 3 and 5 improve sharply ($142.55 \rightarrow 91.57$; $131.31 \rightarrow 98.23$) because deep supervision removes over-refinement. Without it, predictions often stabilize by cycle 12 and drift through cycles 13–20. With it, every cycle is penalized for drift. Fold 1 (the hardest

Table 10: `matbench_phonons`: MAE (cm^{-1}), lower is better. Pre-trained references are shown in a separate block because they use large external DFT corpora.

Model	Method	MAE (cm^{-1})	Params
<i>No pre-training</i>			
DeeperGATGNN	GNN	33.12	–
MODNet v0.1.12	tree-based	34.28	~ 0
MODNet v0.1.10	tree-based	38.75	~ 0
GraphTRIADS V6 (fixed 8 cycles)	Recursive GNN	41.91 ± 4.04	247K
GraphTRIADS V6 (gate-halt 4–12)	Recursive GNN	43.29 ± 4.82	248K
CrabNet [†]	Transformer	55.11	$\sim 600\text{K}$
AMMExpress v2020	RF/AutoML	56.17	–
RF-SCM/Magpie	RF	67.61	–
Dummy (mean)	–	323.76	0
<i>Pre-trained models (not directly comparable)</i>			
MEGNet (pre-trained)	GNN	28.76	$\sim 2\text{M}$
coNGN (pre-trained)	GNN	28.89	$\sim 1\text{M}$
ALIGNN (pre-trained)	GNN	29.54	$\sim 4\text{M}$
coGN (pre-trained)	GNN	29.71	$\sim 1\text{M}$

Table 11: Phonons V6 fold-level comparison: fixed 8 cycles vs. gate-halt (4–12 cycles). Fixed cycles are better on 4/5 folds.

Fold	Fixed 8c	Gate-Halt 4–12c
0	45.06	49.96
1	40.48	38.87
2	37.69	39.06
3	38.76	40.25
4	47.55	48.33
Mean $\pm$ Std	41.91 ± 4.04	43.29 ± 4.82

split, with structurally different alloys) shows minimal gain ($116.32 \rightarrow 118.67$), suggesting irreducible difficulty.

Number of cycles T . Table 13 provides sensitivity analysis of T using available experimental data (same 87K architecture).

Without deep supervision, increasing T from 16 to 20 *hurts* performance (+7.5 MPa average). Hard folds (1, 2, 4) improve modestly with extra cycles (−8.2, −7.0, −6.7 MPa), but easy folds (3, 5) degrade severely (+37.96, +21.53 MPa). Extra cycles over-refine already-correct predictions. With deep supervision at $T = 20$, all folds improve, eliminating over-refinement. We do not have deep-supervised results for $T \in \{8, 12\}$; a full ablation is future work.

4.8 Halting and Deep Supervision Interaction

Table 14 compares halting strategies from V10–V11 experiments.

The dominant gain comes from deep supervision, not halting. ACT ($\lambda = 0.01$, no DS) gives only marginal improvement (−2.0 MPa) and adds a brittle hyperparameter: at $\lambda = 0.01$, ACT collapses to early halting in all folds (mean halt step 15.6/20, 97% early halting), effectively

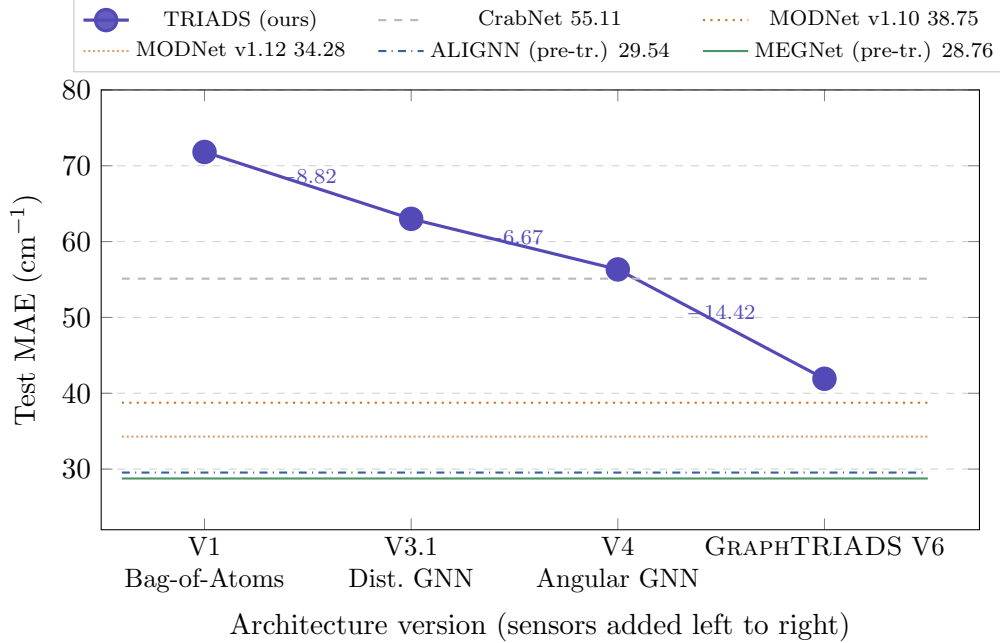


Figure 3: MAE on `matbench_phonons` as physical sensors are added. Each version changes exactly one sensor class; architecture, optimizer, and hyperparameters are held constant. Reference lines denote comparison classes: CrabNet (non-pre-trained neural baseline), MODNet (tree-based), and ALIGNN/MEGNet (pre-trained GNNs, not directly comparable). GRAPHTRIADS V6 crosses below CrabNet and approaches MODNet v1.10 without any pre-training.

Table 12: Deep supervision ablation on `matbench_steels`. Architecture is identical; only the objective differs. Per-fold breakdown included for full transparency.

Config	Mean	Std	F1	F2	F3	F4	F5	Δ
V9A (final-step only)	134.59	10.43	116.32	146.07	142.55	136.70	131.31	—
V10A (deep supervision)	103.29	10.50	118.67	95.32	91.57	112.65	98.23	−23.3%
3-seed check (V10A)	105.85	—	—	—	—	—	—	seed $\sigma=3.00$

shrinking T without improving accuracy. Gate-based halting avoids this by triggering on state convergence instead of penalizing cycle count.

A fixed-steps + DS row is unavailable and is required to cleanly separate DS and halting effects. We treat this as a current limitation.

5 Discussion

5.1 Architecture vs. Featurization: Controlled Evidence

A central question is how much performance comes from architecture versus feature engineering. Our controlled evidence indicates both are necessary: feature quality sets the available signal, while architecture determines how much of that signal is recovered under small-data constraints.

Table 13: Sensitivity to number of reasoning cycles T . V7B and V9A share the same 87K architecture; only T changes. Per-fold breakdown reveals the over-refinement failure mode at $T = 20$ without deep supervision.

Config	T	DS	Mean	F1	F2	F3	F4	F5
V7B (Hybrid-L)	16	No	127.08	124.56	153.03	104.59	143.42	109.78
V9A (Hybrid-20)	20	No	134.59	116.32	146.07	142.55	136.70	131.31
V10A (+ DS)	20	Yes	103.29	118.67	95.32	91.57	112.65	98.23

Bold values highlight the over-refinement failure: easy folds 3 and 5 degrade badly (+37.96, +21.53 MPa) when T increases from 16 to 20 without deep supervision, then recover with DS.

Table 14: Halting mechanisms on `matbench_steels`. All rows: same base architecture, 20 cycles. Gate-halt row uses deep supervision; ACT row does not (this confound is noted).

Halting method	DS	MAE (MPa)	Params
Fixed 20 steps, no DS	No	134.59	87K
ACT ($\lambda=0.01$, no DS)	No	132.59	89K
Gate-based halt + DS	Yes	103.29	87K

Supporting featurization experiment on `matbench_expt_gap`. Table 15 holds the architecture constant and changes only the input features.

Table 15: Controlled featurization test on `matbench_expt_gap`: same architecture, different inputs. Both HYBRIDTRIADS rows use the same model hyperparameters; only the feature set changes.

Model		Feature set	Dim	MAE (eV)	Params
HYBRIDTRIADS (generic) [†]	V1	Magpie + ElementFraction + Mat2Vec	468d	0.3616	82K
HYBRIDTRIADS (physics-informed) [†]	V2	Magpie + BandCenter + HOMO/LUMO + Mat2Vec	354d	0.3342	75K
HYBRIDTRIADS V3 (5-seed, 100K) [†]	V3	Same as V2, scaled	354d	0.3068	100K
CrabNet (ref.) [†]		Composition tokens	–	0.3463	~600K

[†] Composition-only. BandCenter and HOMO/LUMO computed from NIST orbital energies.

V1 $\rightarrow$ V2 yields a 7.6% MAE reduction by changing *only* input features, with no model-code change. V1 (generic features) is close to CrabNet (0.3616 vs. 0.3463), while V3 (same core architecture with physics-informed features and scaling) reaches 0.3068. This gain is meaningful but smaller than the 23.3% architecture-only improvement from deep supervision on `matbench_steels`.

Architecture experiment on `matbench_steels`. The version history (Table 3) isolates architectural impact. The key step is V9A $\rightarrow$ V10A: architecture and features are identical; only the objective changes (deep supervision vs. final-step loss). This yields a 23.3% MAE reduction, the largest single-step gain in steels development. The same table also shows why this gain is not a generic training trick: the attention-cell transition (V7B) is modest on its own, but it sets up the large deep-supervision gain at V10A.

Synthesis. The $V9A \rightarrow V10A$ ablation is clean controlled evidence: architecture and features are identical, and changing only the objective (final-step loss $\rightarrow$ per-cycle supervision) reduces MAE from 134.59 to 103.29 MPa. This shows that architecture-coupled training dynamics can deliver a large gain even when features are fixed. Conversely, the `matbench_expt_gap` feature control ($V1 \rightarrow V2$) shows that sensor changes alone also produce meaningful improvement (7.6%).

Architectural contribution. The core finding is direct: TRIADS’ parameter efficiency comes from the coupled design, not from any single ingredient. Physics-informed sensors provide usable priors; shared attention-based recursion turns those priors into relational multi-step updates; per-cycle supervision forces every step to stay prediction-calibrated. The controlled results support this claim: removing deep supervision sends steels MAE from 103.29 back to 134.59 MPa, and weakening structural sensors keeps phonons at 71.82 cm^{-1} until geometric information is added. The contribution is this coupled mechanism and its measured effect.

5.2 Deep Supervision as a Small-Data Training Principle

The deep supervision result on `matbench_steels` is the central empirical finding: with identical architecture and sensor set, per-cycle supervision reduces MAE from 134.59 to 103.29 MPa (23.3% relative). This is direct evidence, within TRIADS, of an architecture-coupled training effect consistent with the attention-based cell producing meaningful intermediate states throughout recursion.

The over-refinement pattern (Table 12, fold 3: 142.55 without DS vs. 91.57 with DS) is a concrete failure mode. A plausible mechanism is that once a final-step-only model reaches a good prediction by cycle k , later cycles can destabilize it because shared weights are still optimizing harder folds; easy-fold predictions become collateral damage. Deep supervision keeps every cycle calibrated and suppresses this drift. This is inferred from fold-level outcomes; direct confirmation requires per-cycle training logs.

5.3 Limitations

Phonons gap to pre-trained GNNs. The 13.15 cm^{-1} gap between GRAPHTRIADS (41.91) and MEGNet (28.76) likely reflects pre-training data scale more than architecture. A pre-trained GRAPHTRIADS checkpoint could narrow this gap.

JDFT2D fold heterogeneity. The 5-seed ensemble improves mean MAE, but variance remains high ($\pm 12.40 \text{ meV/atom}$) because fold 2 is consistently difficult across seeds (54–63 meV/atom).

Single-seed metallicity. `matbench_expt_is_metal` is reported from one seed; fold variance (± 0.0029) is low, and additional seeds are expected to reduce variance but were omitted due to compute constraints.

Gate regularizer sensitivity. All reported runs use $\gamma = 0.001$ in Eq. (5); we do not yet provide a systematic γ sweep.

Incomplete T -ablation. We have $T = 16$ (no DS) and $T = 20$ (no DS and with DS) on `matbench_steels`. A full $T \in \{8, 12, 16, 20\}$ ablation with deep supervision is not available.

Coverage of remaining Matbench tasks. Evaluation on the seven unevaluated Matbench tasks was deferred due to compute constraints.

ACT/DS confound. Table 14 compares ACT (no DS) to gate-halt (with DS). A fair halting comparison requires fixed+DS vs. gate-halt+DS.

SWA contribution uncontrolled. Stochastic Weight Averaging [11] is used in all TRIADS experiments, but its effect is not isolated because the main results lack no-SWA baselines. An early steels run (V5A with SWA: 128.98 MPa vs. prior best without: 130.33 MPa) suggests a modest improvement. Future work should include paired SWA/no-SWA runs.

5.4 Negative Result: HTRM

The Hierarchical TRM (HTRM, V15) was built following the HRM design [8]: a high-level attention module ($n_H = 4$ cycles) with gradient detachment provides abstract context to a low-level MLP-TRM module ($n_L = 5$ steps). On `matbench_steels`, HTRM degraded to MAE 431.87 ± 49.59 MPa with per-fold results [367.70, 412.33, 422.00, 519.40, 437.91], worse than the dummy mean predictor (229.74 MPa). Predictions are worse than the unconditional mean, indicating the model learned no useful signal.

This failure likely comes from gradient detachment: with only ≈ 200 samples per fold, the high-level module receives no gradient from final loss, and the low-level module cannot compensate alone. HRM-style designs succeed in text domains with dense supervision from massive corpora; in small-data materials settings, detachment is catastrophic.

6 Summary

Table 16: TRIADS results across six Matbench benchmarks. Values are manuscript-archived metrics; external leaderboards may update over time.

Task	Metric	TRIADS	Params	Seeds
<code>matbench_steels</code>	MAE↓	91.20 ± 12.23 MPa	225K	5
<code>matbench_expt_gap</code>	MAE↓	0.3068 ± 0.0082 eV	100K	5
<code>matbench_expt_is_metal</code>	AUC↑	0.9655 ± 0.0029	100K	1
<code>matbench_glass</code>	AUC↑	0.9285 ± 0.0063	44K	5
<code>matbench_jdft2d</code>	MAE↓	35.89 ± 12.40 meV/atom	75K	5
<code>matbench_phonons</code>	MAE↓	41.91 ± 4.04 cm ⁻¹	247K	1

7 Conclusion

We introduced TRIADS, a family of recursive architectures for materials property prediction evaluated on six Matbench benchmarks. The two variants, HYBRIDTRIADS for compositional tasks and GRAPHTRIADS for structural tasks, share weight-tied reasoning cycles, GRU-style gated residuals, per-cycle deep supervision, and implicit gate-based halting. These choices target the small-data regime typical in materials science (312–5,680 samples here).

Key results include ROC-AUC 0.9655 on metallicity classification (+4.46 points over AMMExpress in the March 2026 leaderboard snapshot), MAE 0.3068 eV on `matbench_expt_gap` among confirmed composition-only models, MAE 35.89 ± 12.40 meV/atom on `matbench_jdft2d` with a 5-seed ensemble, and MAE 41.91 ± 4.04 cm⁻¹ on `matbench_phonons` with a 247K-parameter model trained from scratch. On `matbench_jdft2d`, TRIADS outperforms pre-trained coGN/coNGN while using only in-fold Matbench data and a 75K-parameter model.

Table 3 clarifies the architectural mechanism: the attention-based recursive cell (V7B) provides modest standalone gains, but enables the large deep-supervision jump at V10A

when recursion is extended. The central takeaway is that shared attention-based recursive depth, trained with per-cycle supervision and paired with physics-informed sensors, extracts more signal per parameter than the comparably sized fixed-depth baselines tested here, from composition-only prediction to structural-property prediction.

Evaluation on the remaining Matbench tasks is ongoing and was deferred in this submission cycle due to compute constraints.

Code, training logs with fold-level statistics, model checkpoints, and experimental data (including the HTRM failure) are released at:

<https://github.com/Rtx09x/TRIADS>
<https://huggingface.co/Rtx09/TRIADS>
<https://rtx09x.github.io>

Disclosure. Paper write-up/editing and parts of code authoring were AI-augmented. All experiments, numerical checks, and final scientific claims were reviewed and validated by the author.

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